

# Simplifying Matching Methods for Causal Inference

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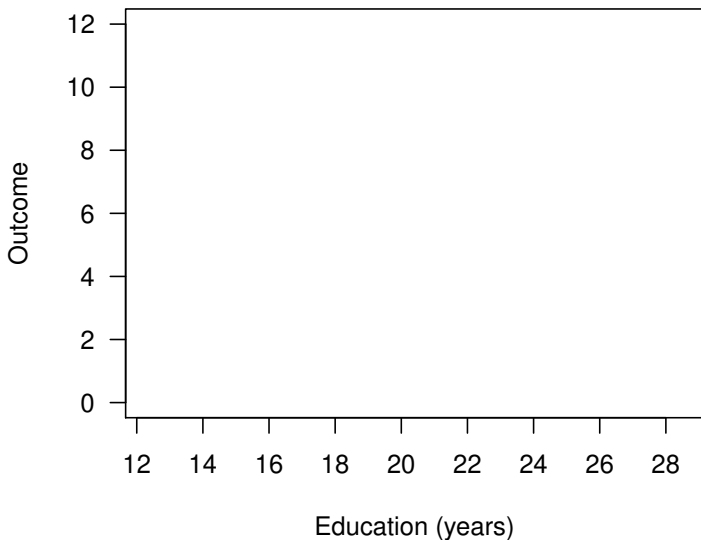
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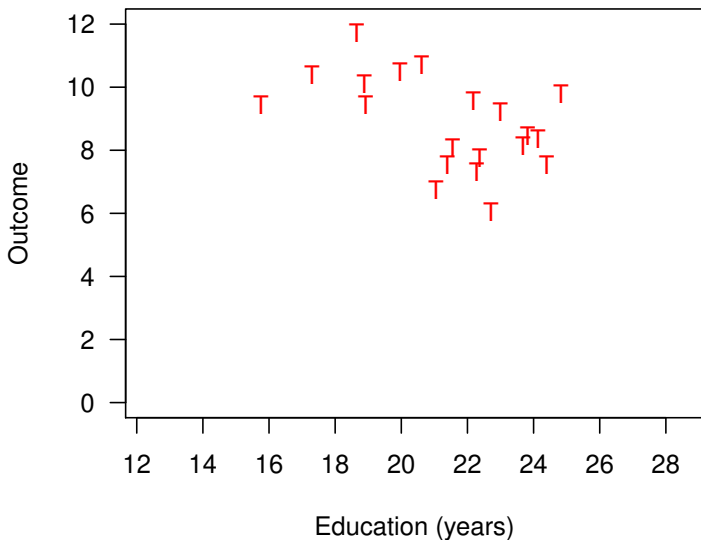
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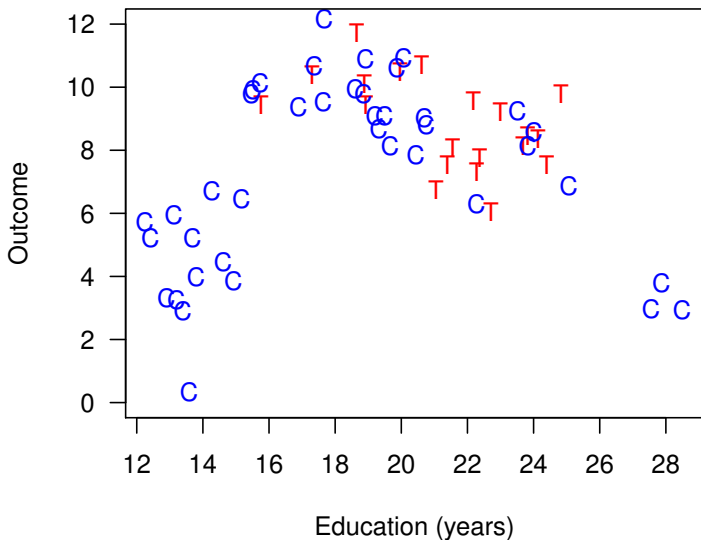
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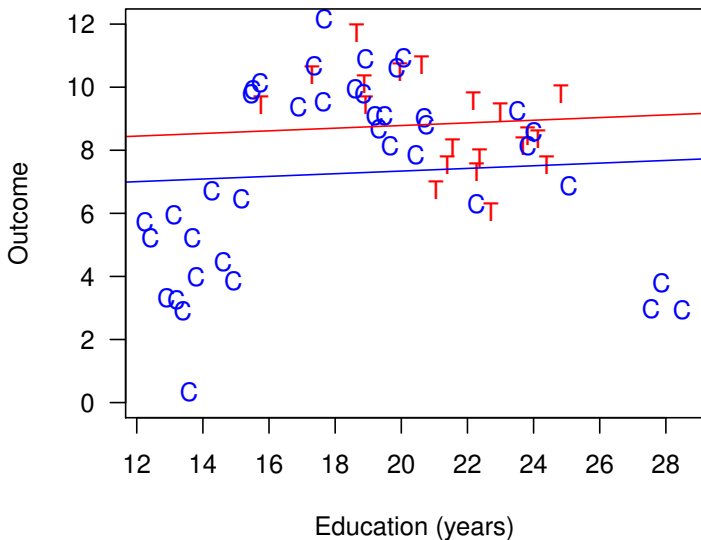
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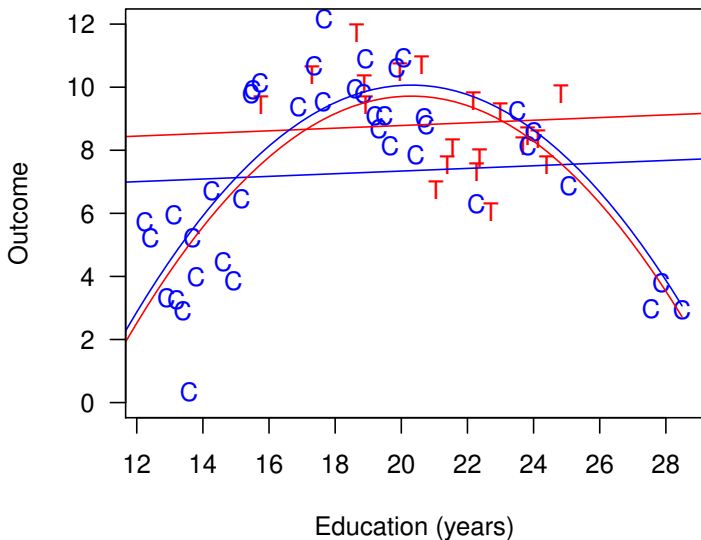
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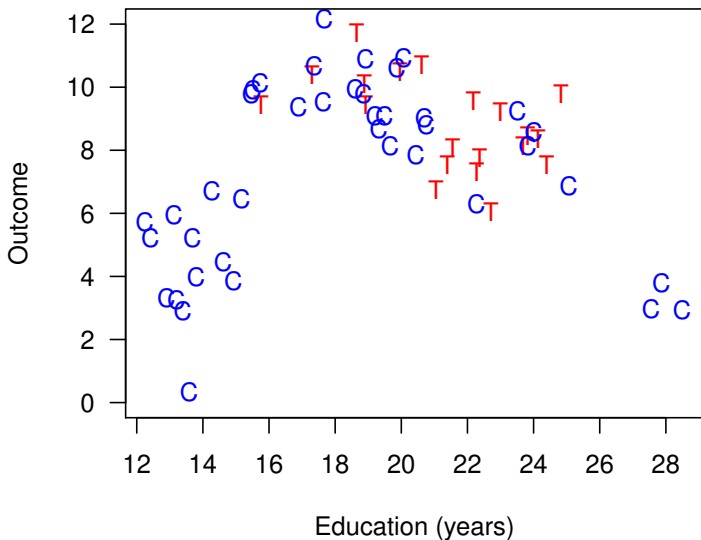
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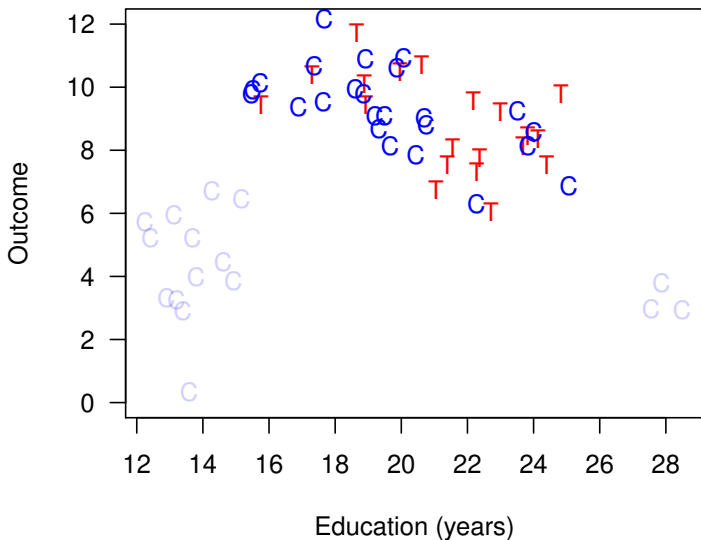
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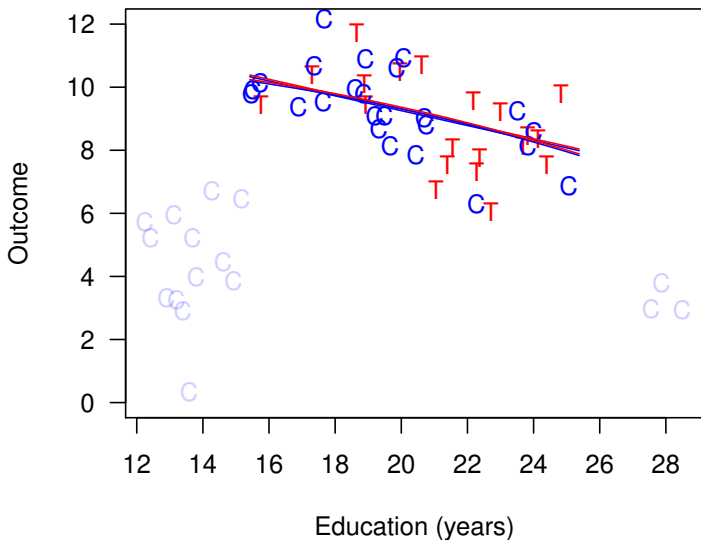
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- Big convenience: Follow preprocessing with whatever statistical method you'd have used without matching

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- Easy extensions for: multi-level, continuous, & mismeasured treatments;  $A$  too wide,  $n$  too small

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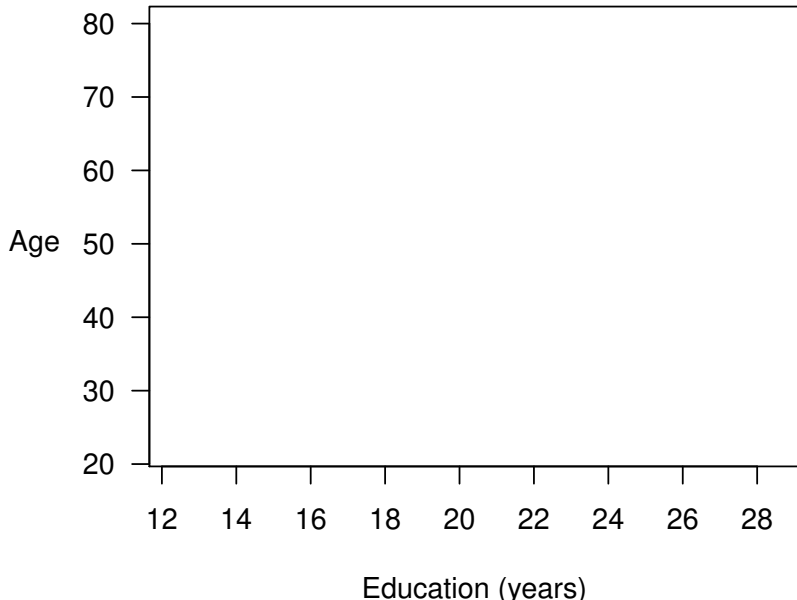
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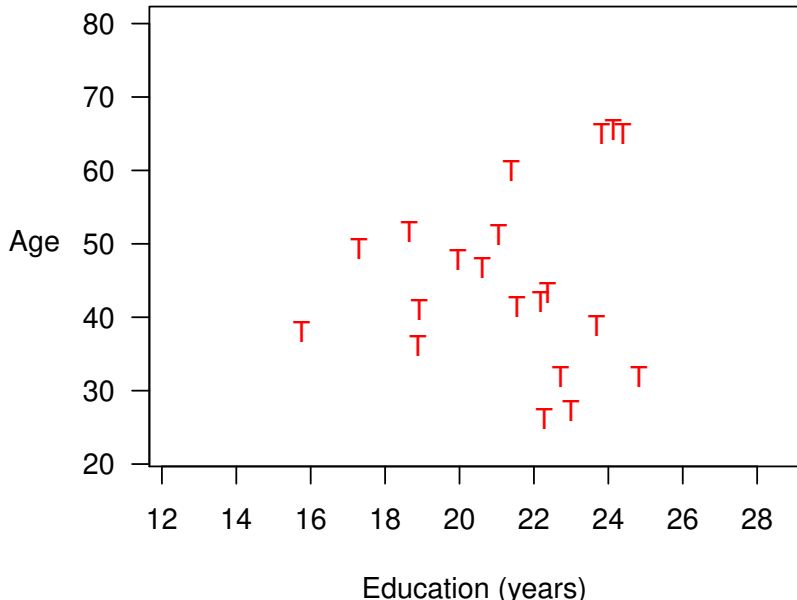
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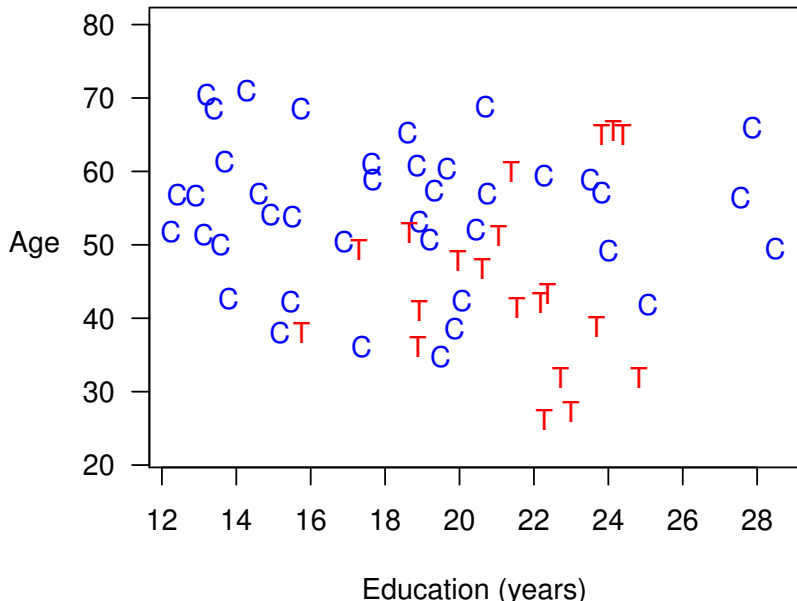
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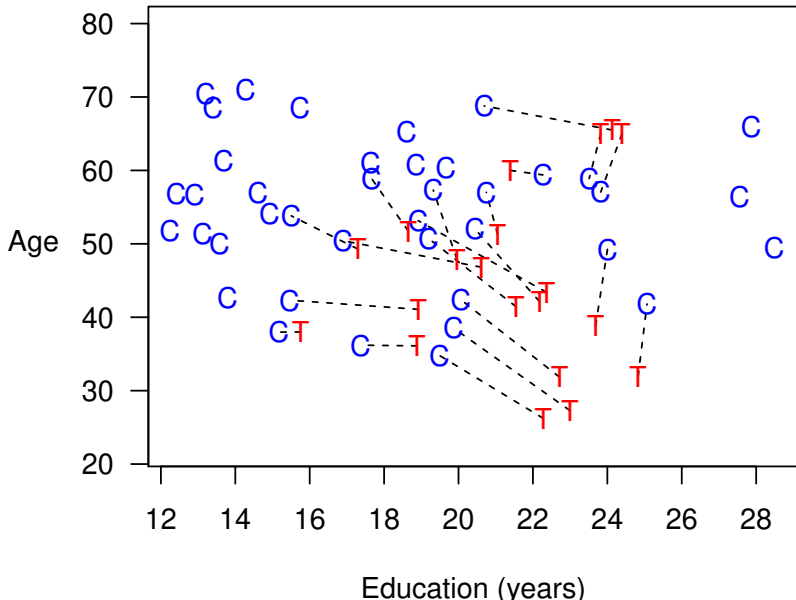
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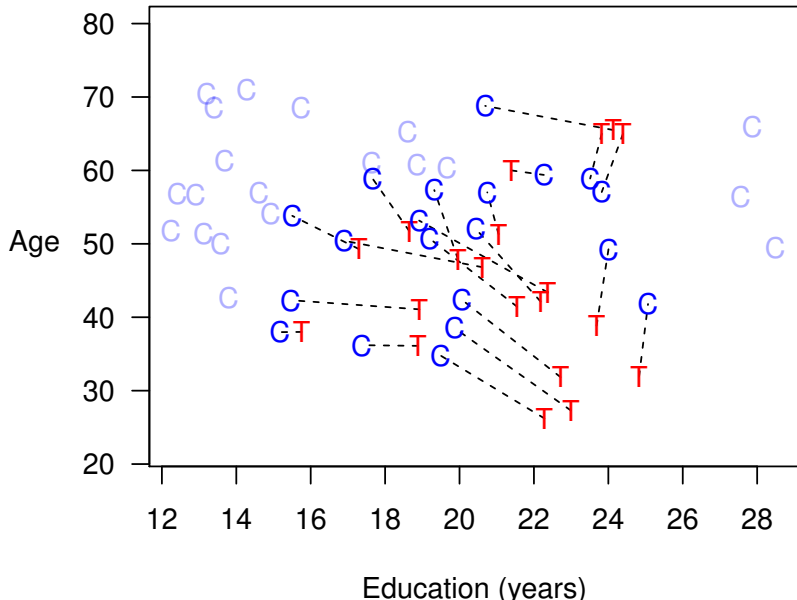
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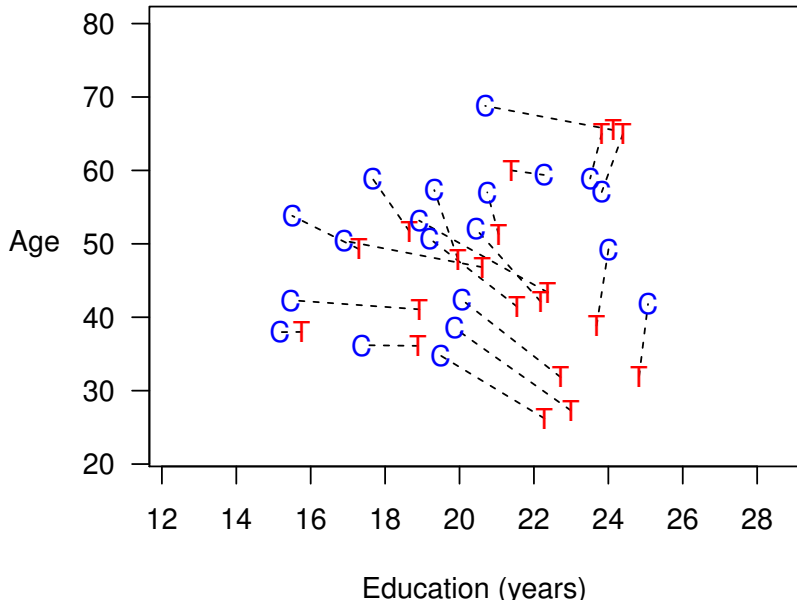
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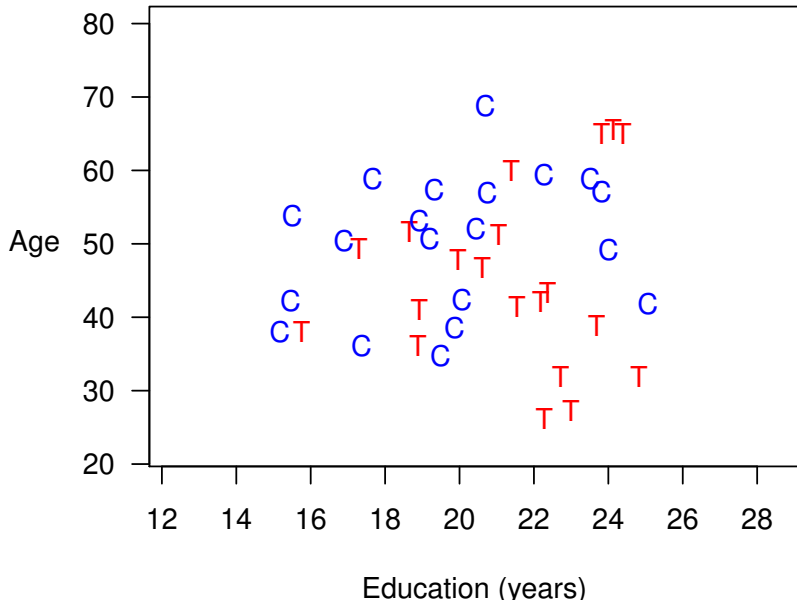
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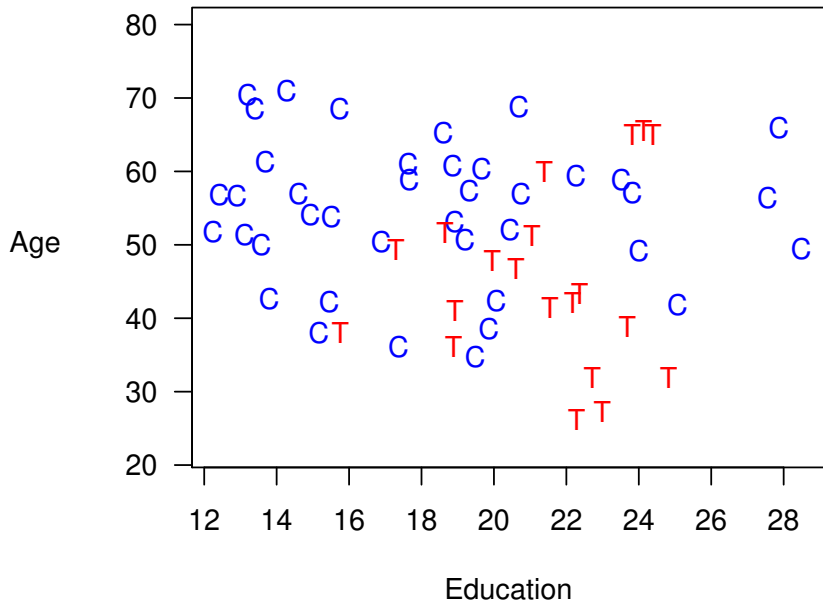
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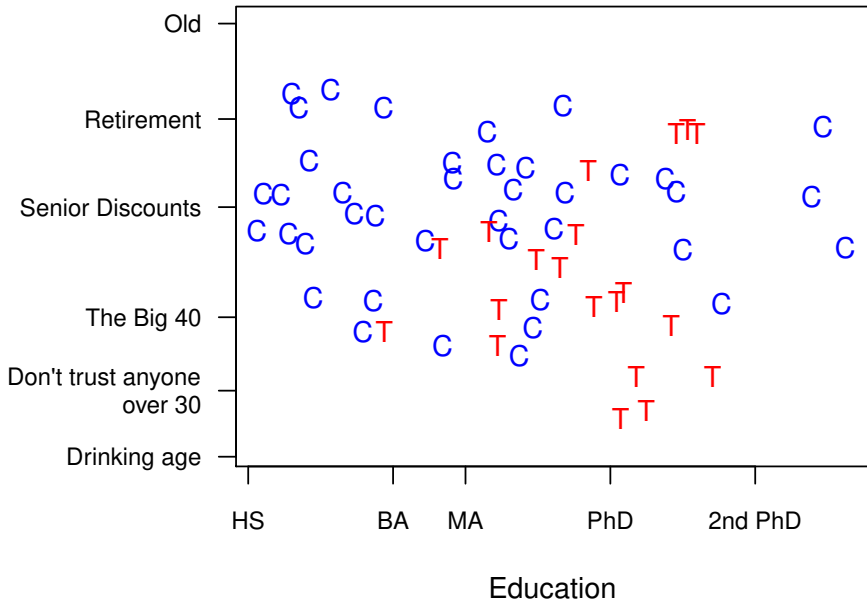
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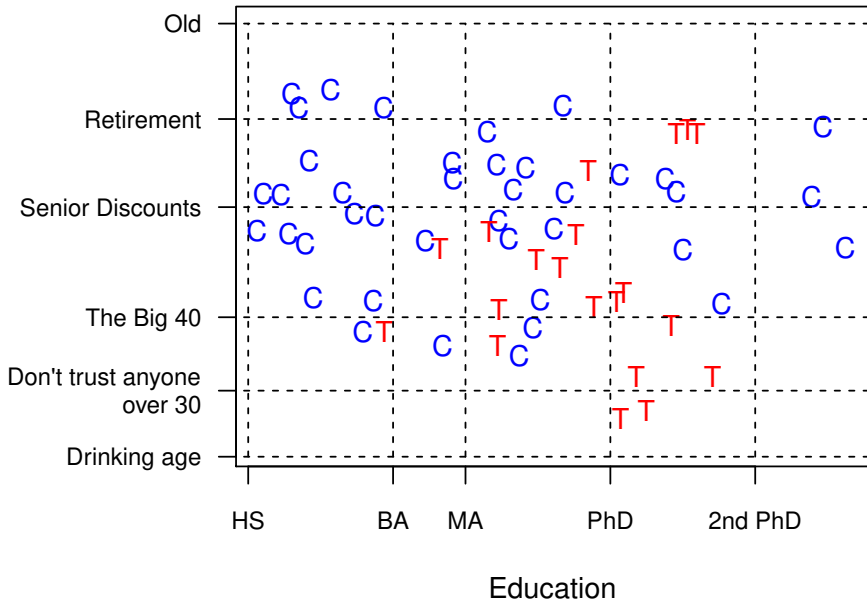
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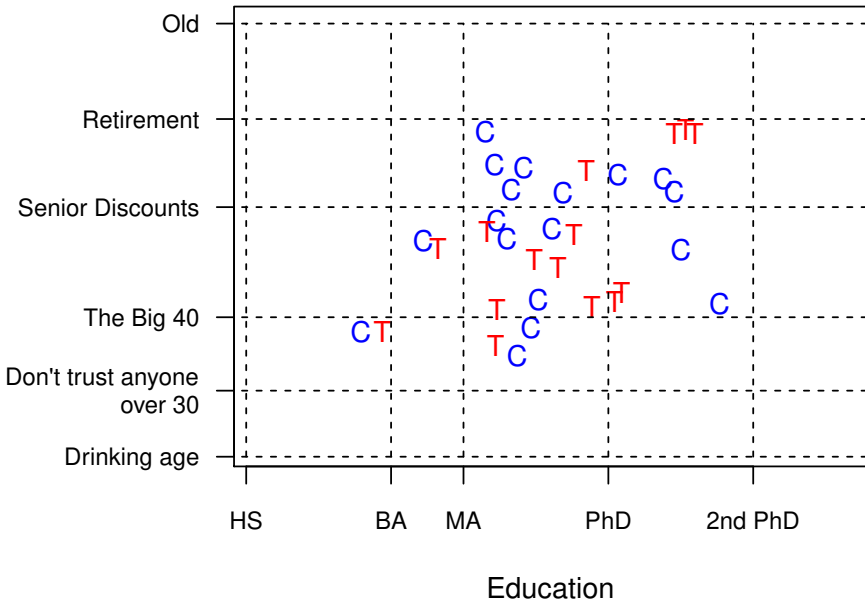
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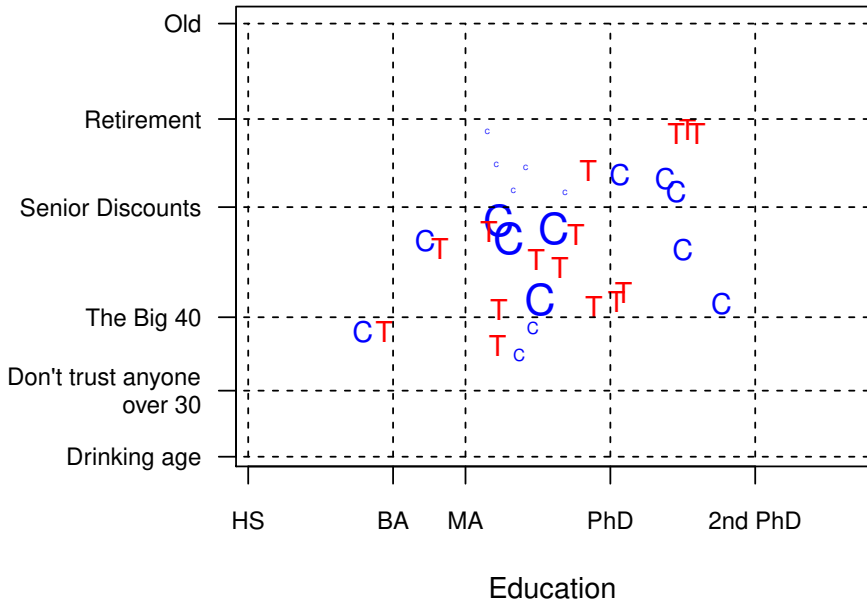
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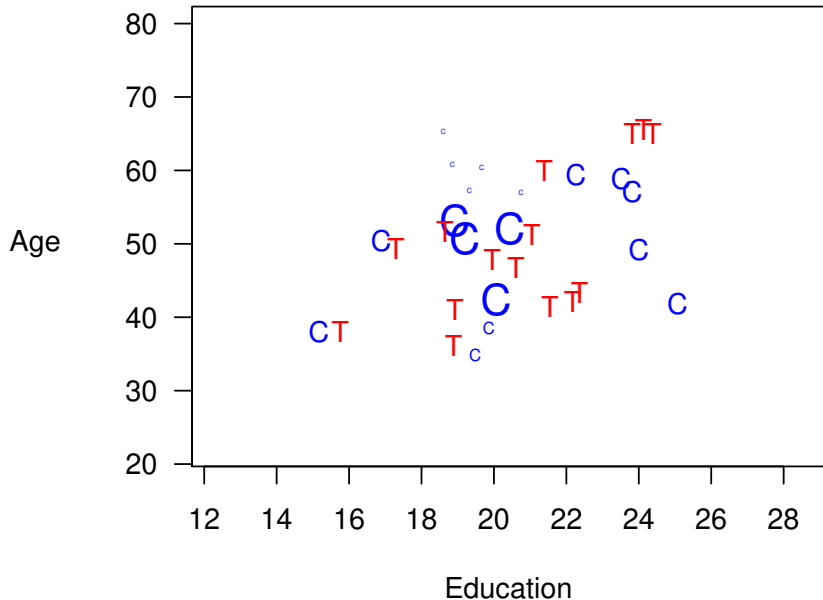
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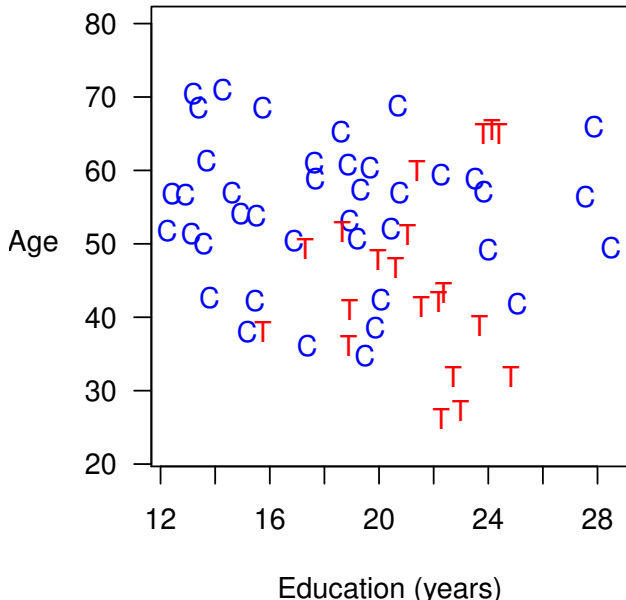
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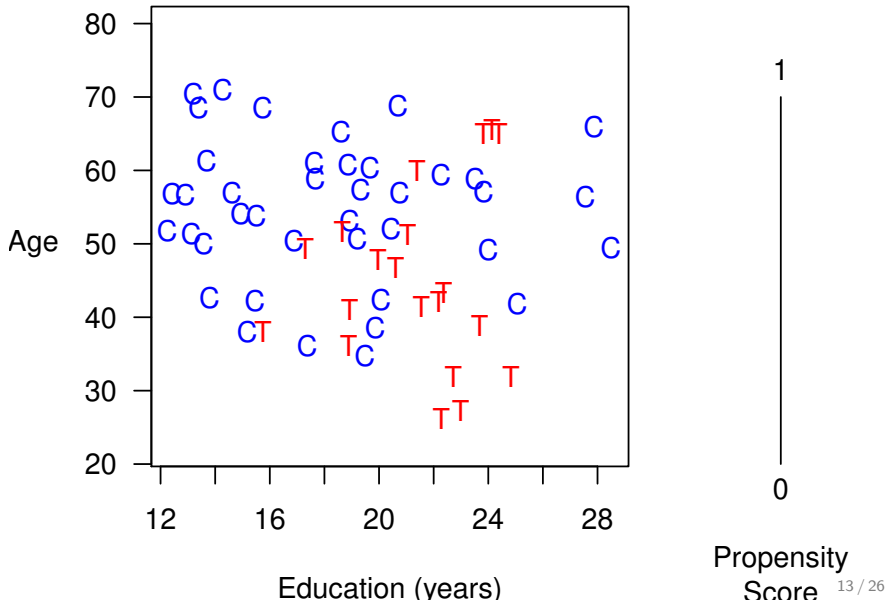
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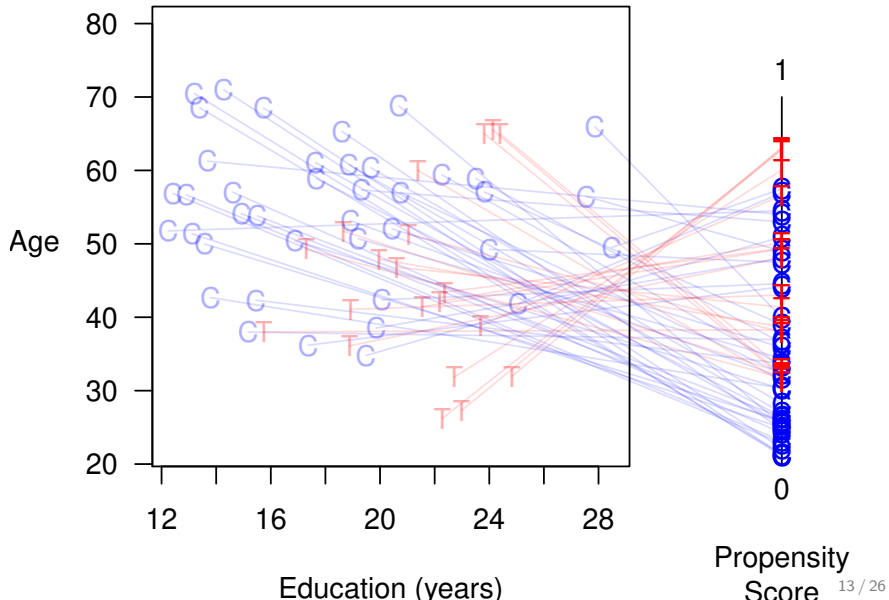
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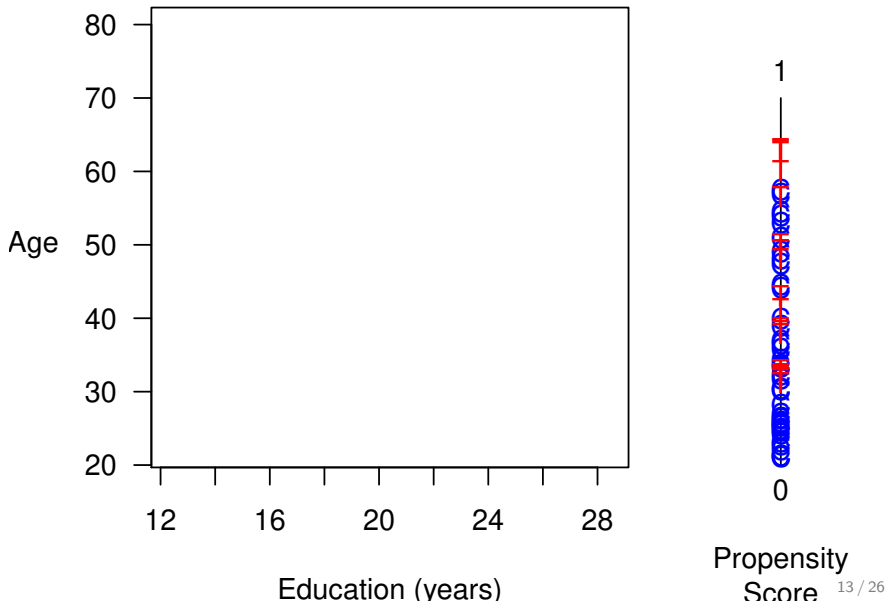
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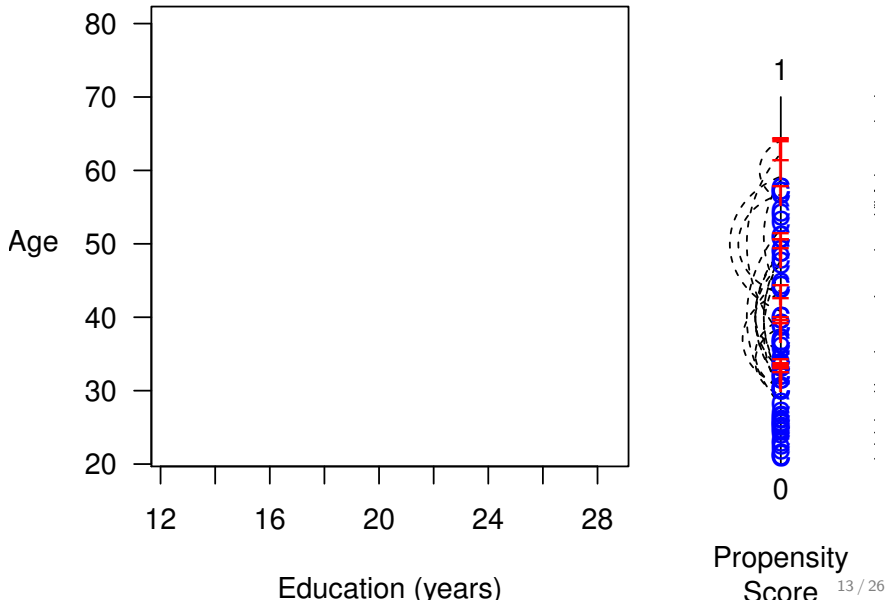
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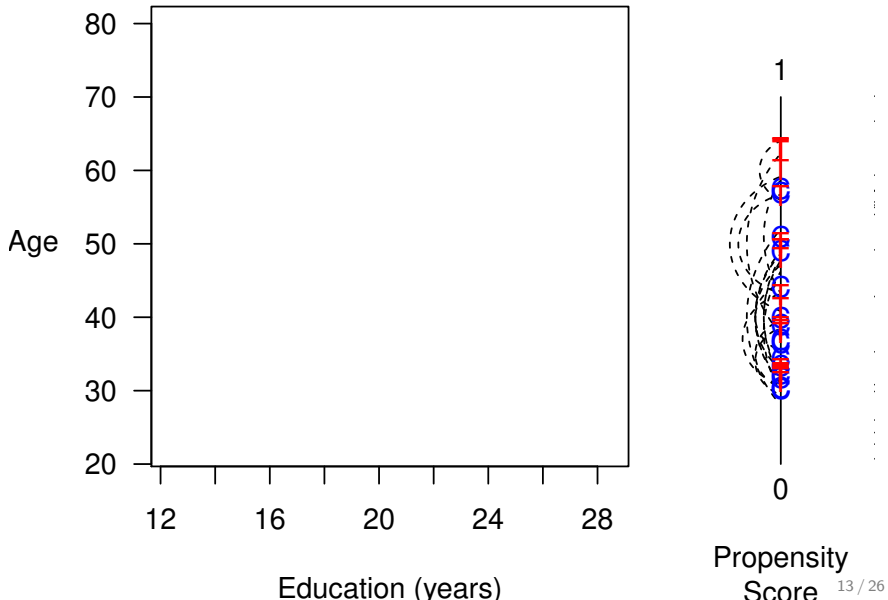
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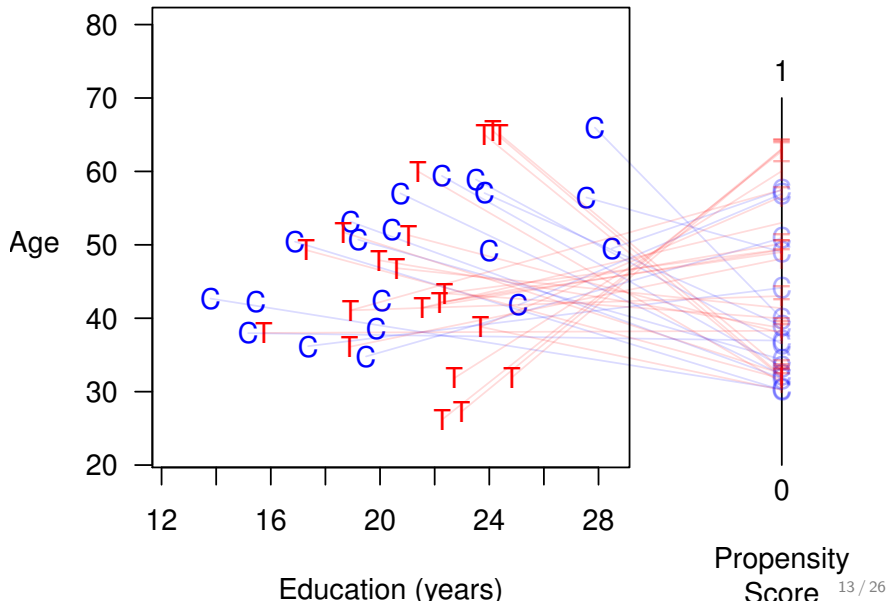
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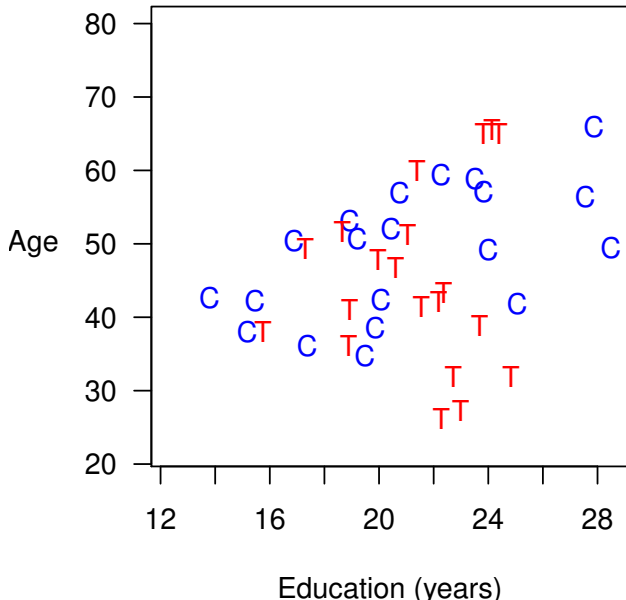
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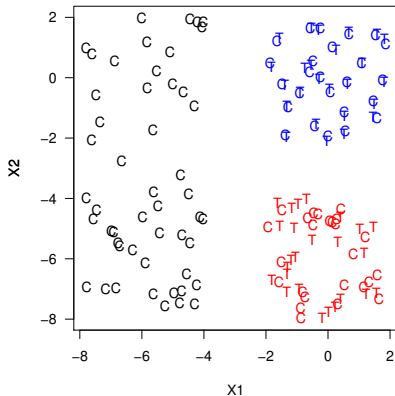
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# PSM's Statistical Properties

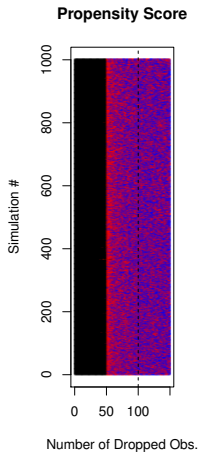
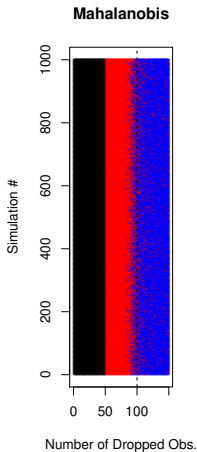
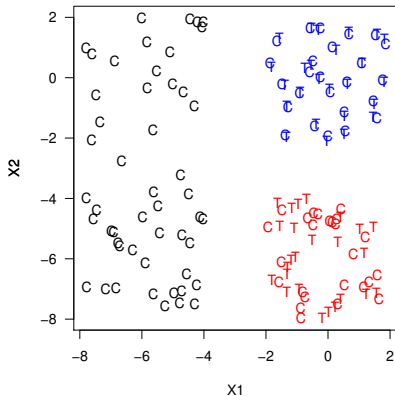
- **PSM is Inefficient:**
  - Efficient relative to complete randomization, but
  - Inefficient relative to full blocking (Imai, King, and Nall: up to 600% difference in SEs in experiments)
- **The PSM Paradox:**
  - If data are balanced to begin with, or after some pruning,  $\hat{\pi} \approx 0.5$  (or constant within strata)  $\rightsquigarrow$  matching is at random
  - Random matching increases imbalance!
  - Approximating complete randomization (by pruning)  $\rightsquigarrow$  higher imbalance  $\rightsquigarrow$  more inefficiency
  - If the data have no good matches, the paradox won't be a problem but you're cooked anyway
- **PSM is Biased:**
  - Imbalance  $\rightsquigarrow$  Inefficiency  $\rightsquigarrow$  Model dependence  $\rightsquigarrow$  Bias
- **Curse of Dimensionality Problems:**
  - The Promise: avoid it by balancing on  $\pi$  rather than  $X$
  - The Reality: The PSM Paradox is bigger with more covariates

# PSM is Blind Where Other Methods Can See

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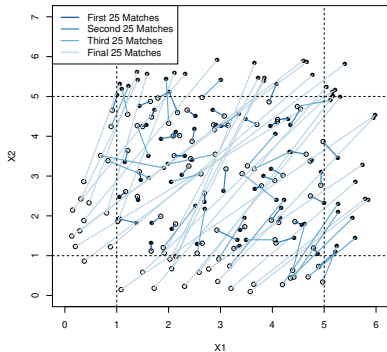


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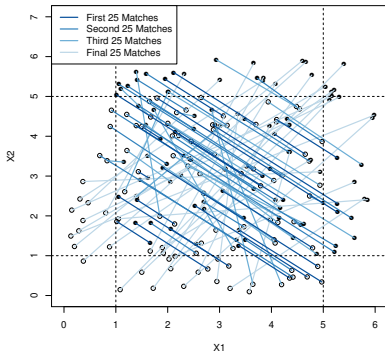


# What Does PSM Match?

## MDM Matches



## PSM Matches

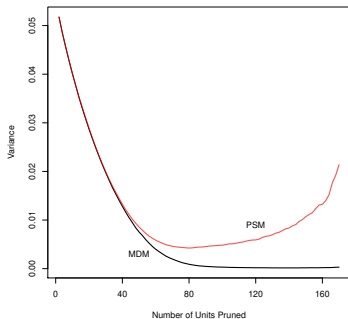


Controls:  $X_1, X_2 \sim \text{Uniform}(0,5)$

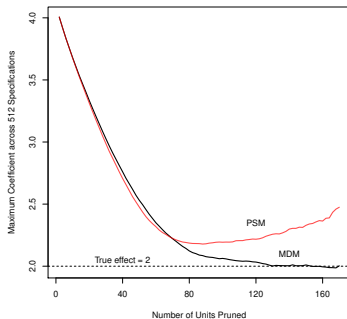
Treateds:  $X_1, X_2 \sim \text{Uniform}(1,6)$

# PSM Increases Model Dependence & Bias

## Model Dependence



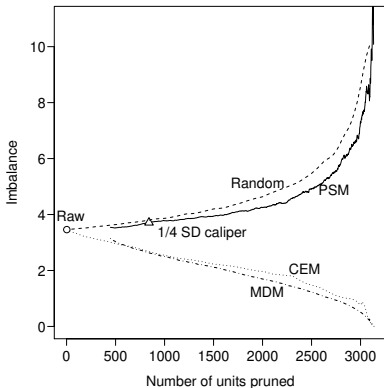
## Bias



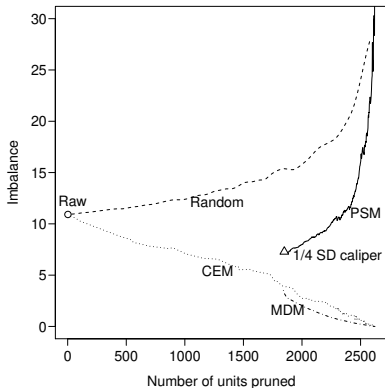
$$Y_i = 2T_i + X_{1i} + X_{2i} + \epsilon_i$$
$$\epsilon_i \sim N(0, 1)$$

# The Propensity Score Paradox

Finkle et al. (2012)



Nielsen et al. (2011)



# The Matching Frontier

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  - No cherry picking possible; you see everything optimal

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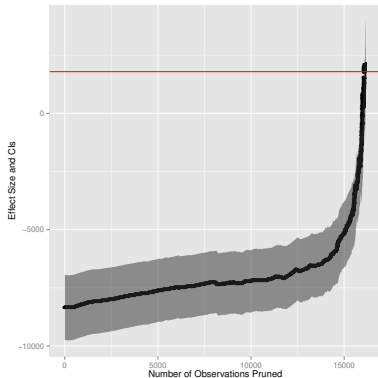
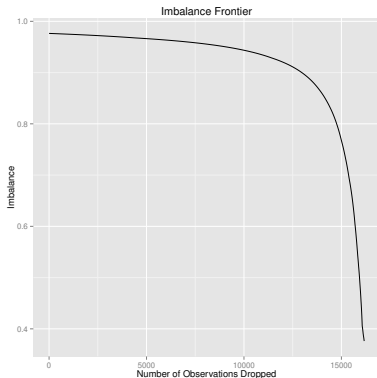
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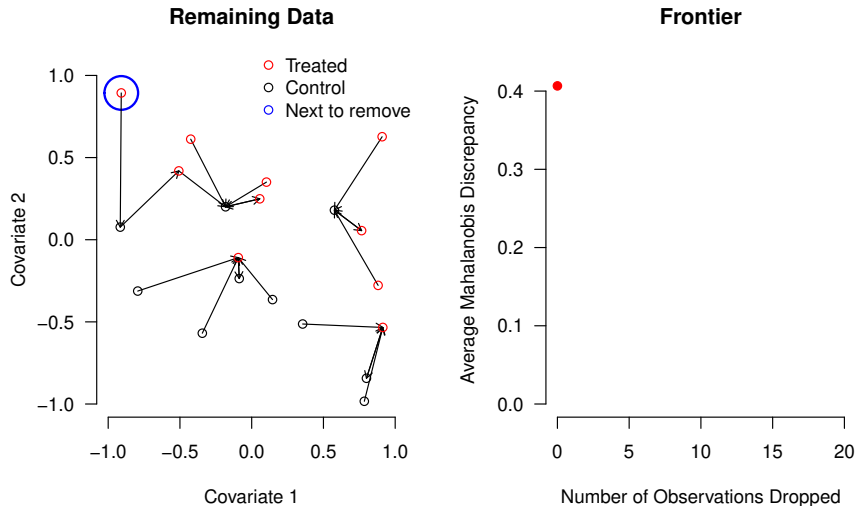
# Job Training Data: Frontier and Causal Estimates



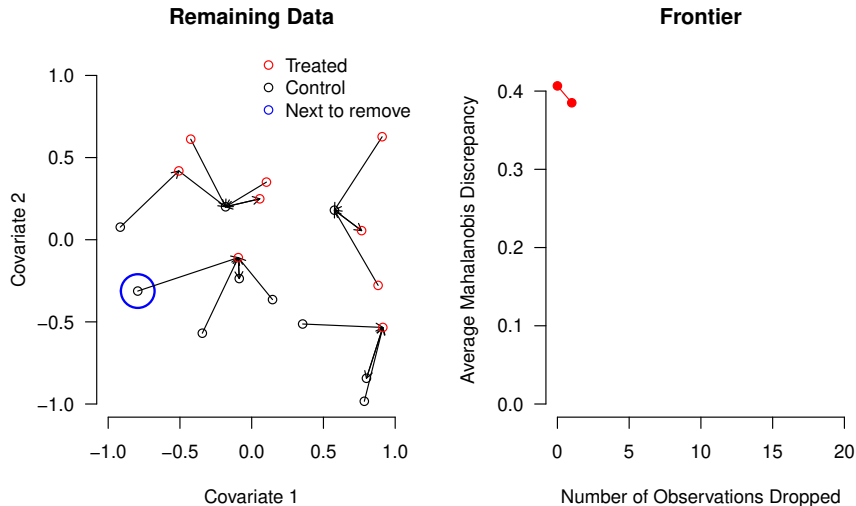
- 185 Ts; pruning most 16,252 Cs won't increase variance much
- Huge bias-variance trade-off after pruning most Cs
- Estimates converge to experiment after removing bias
- No mysteries: basis of inference clearly revealed

# Constructing the FSATT Mahalanobis Frontier

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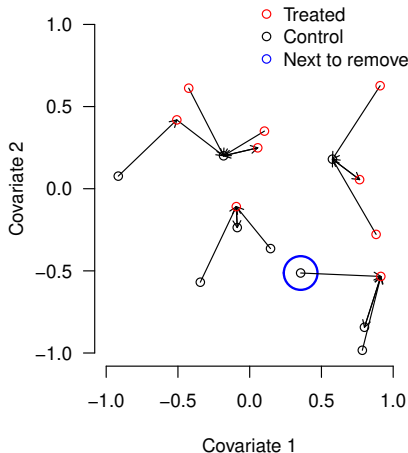


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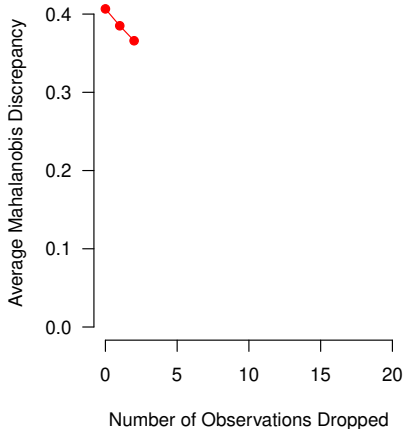


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Remaining Data

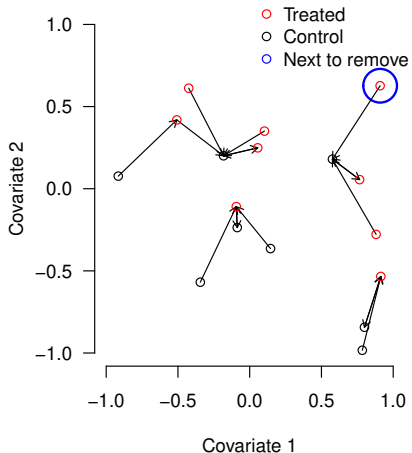


Frontier

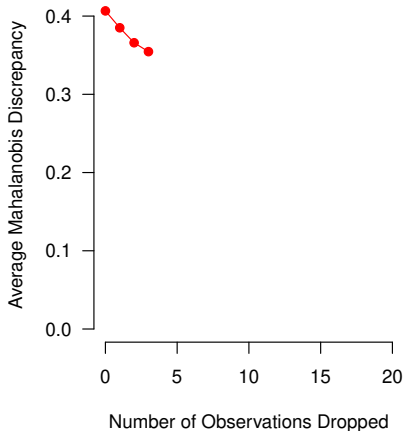


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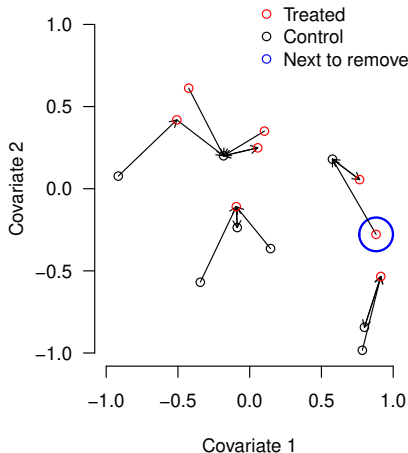


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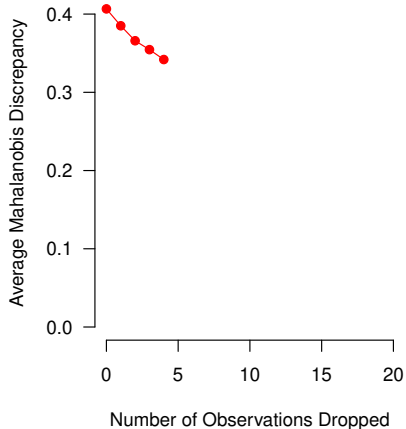


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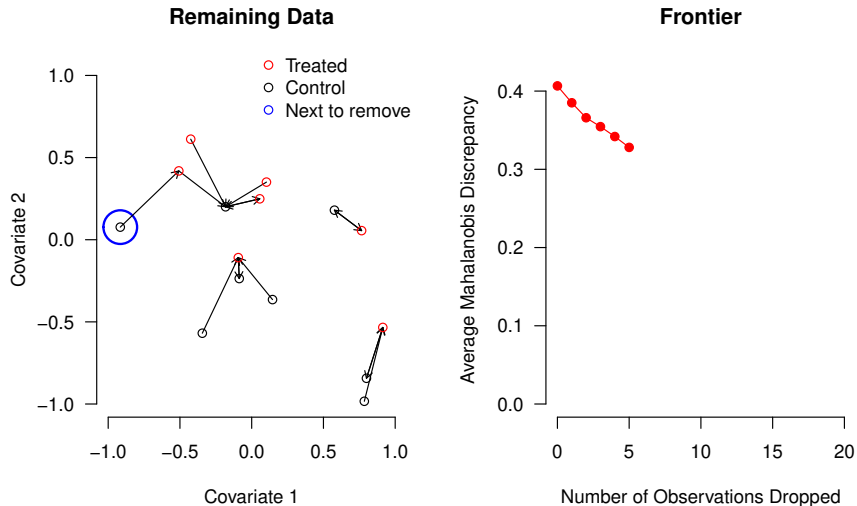
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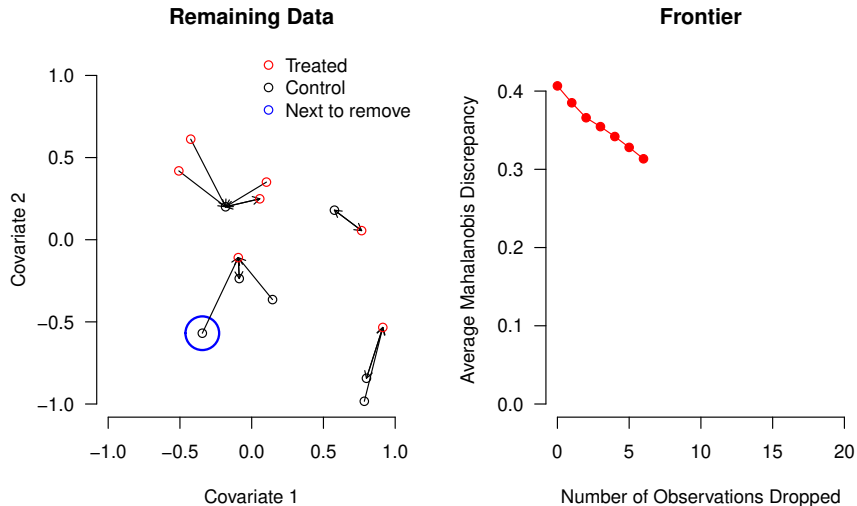
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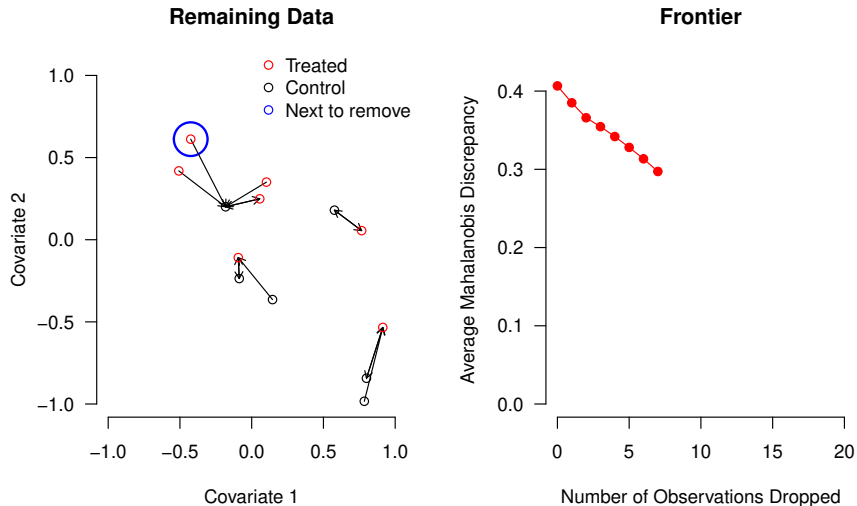
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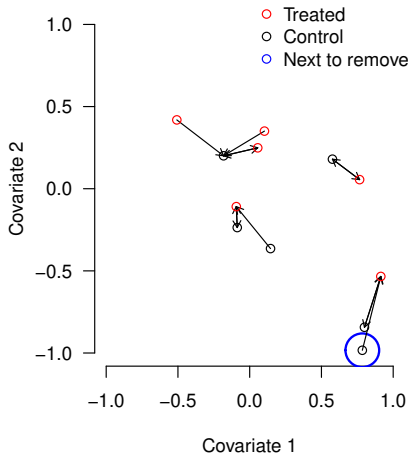


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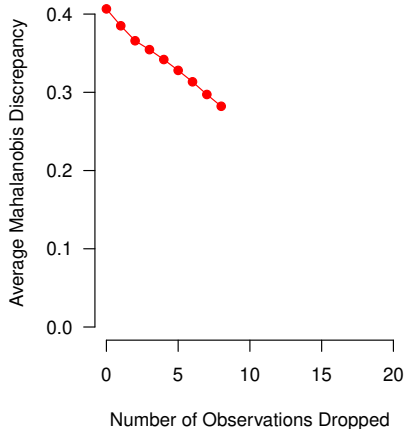


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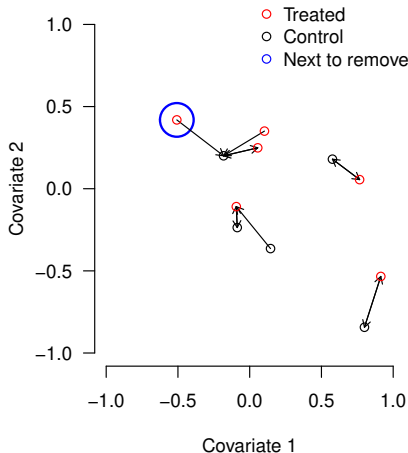


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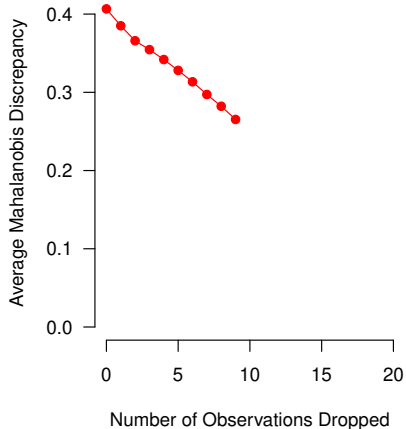


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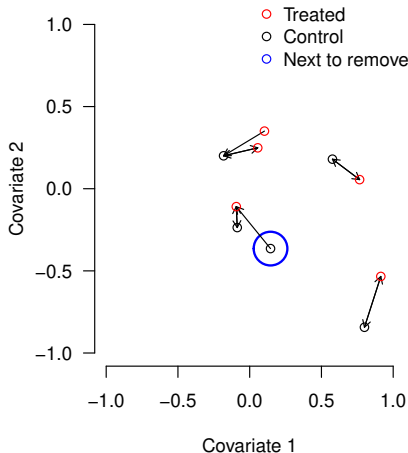


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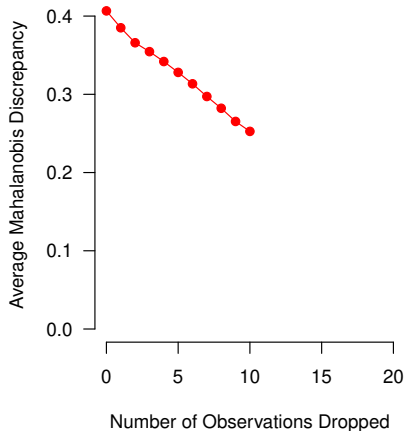


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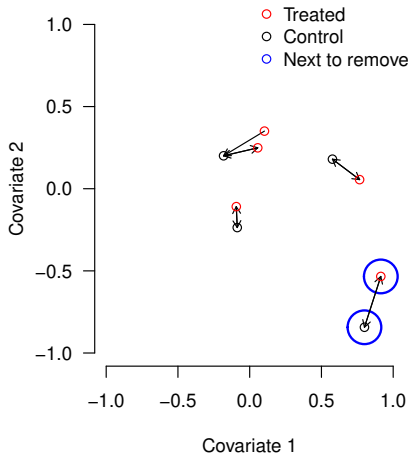


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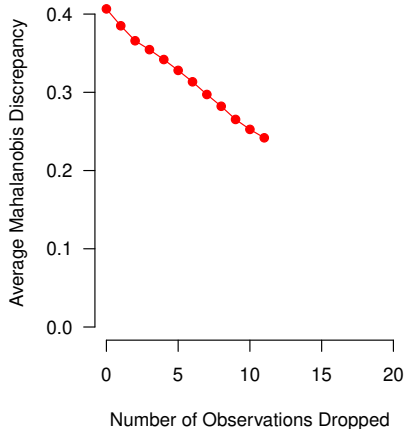


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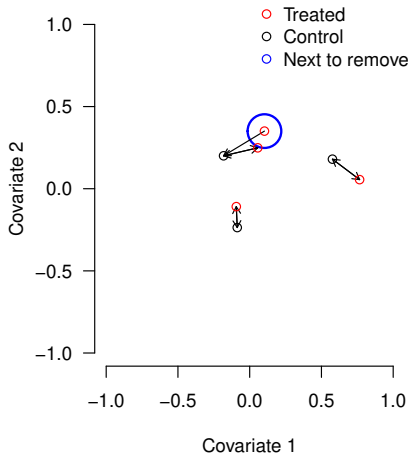


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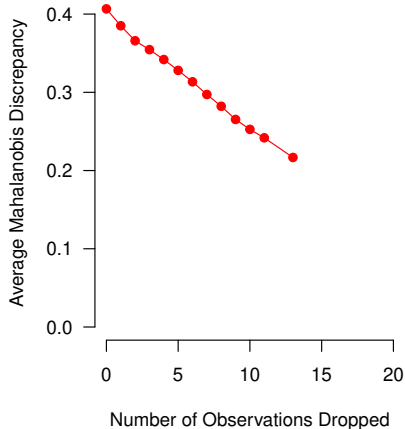


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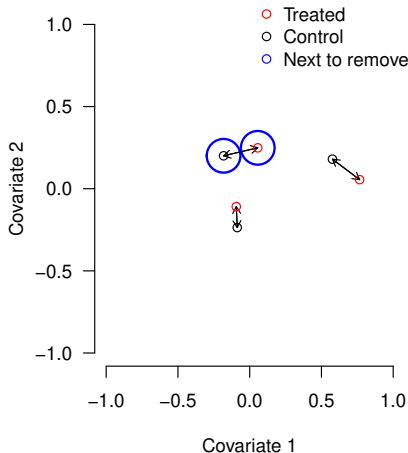


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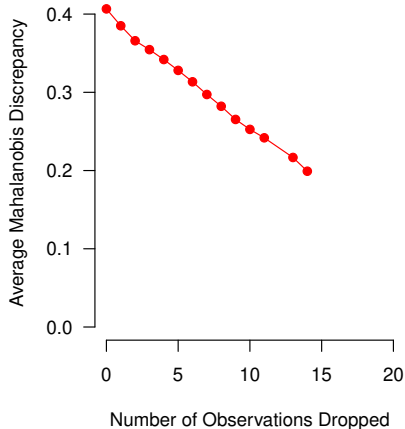


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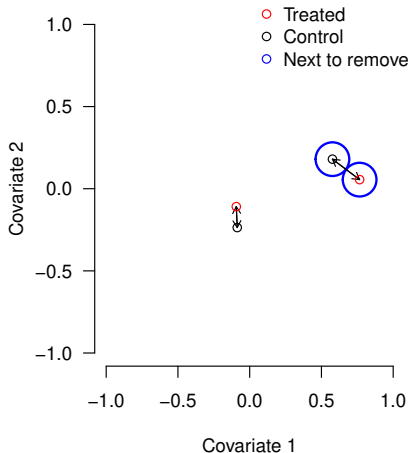


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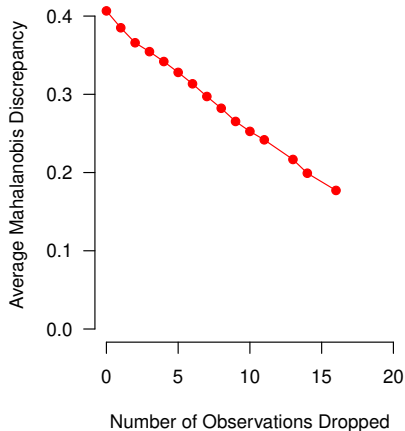


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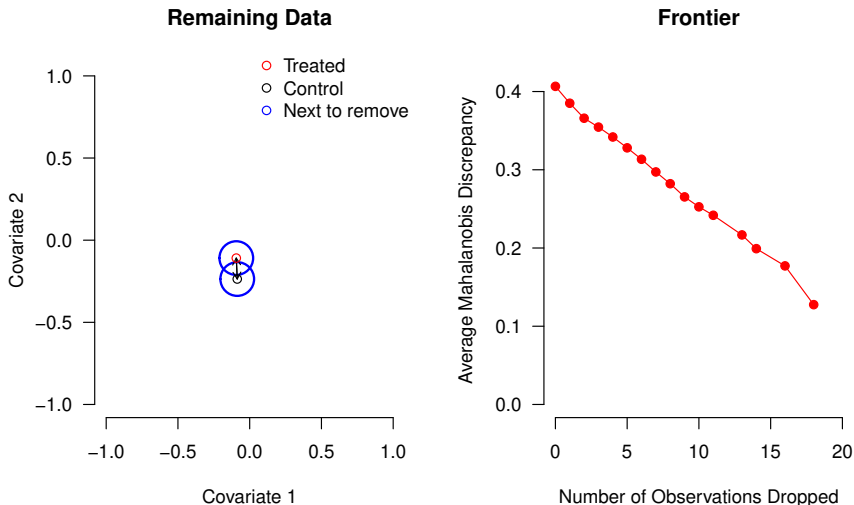
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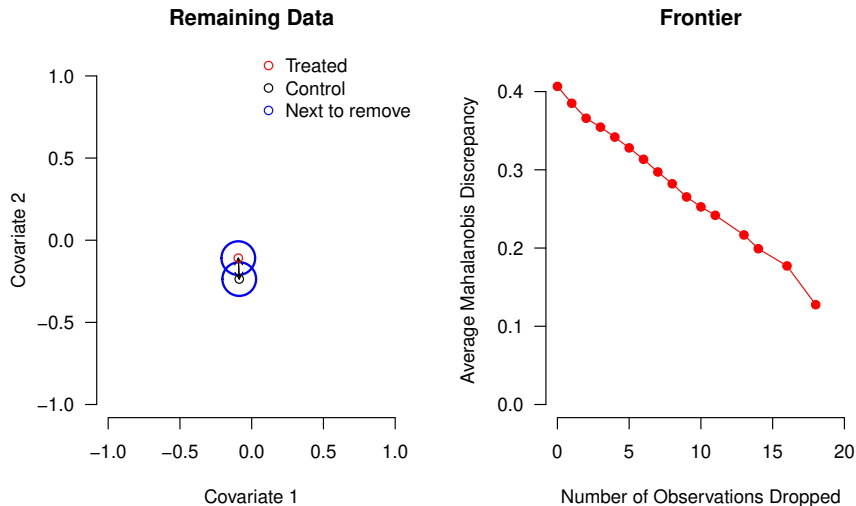
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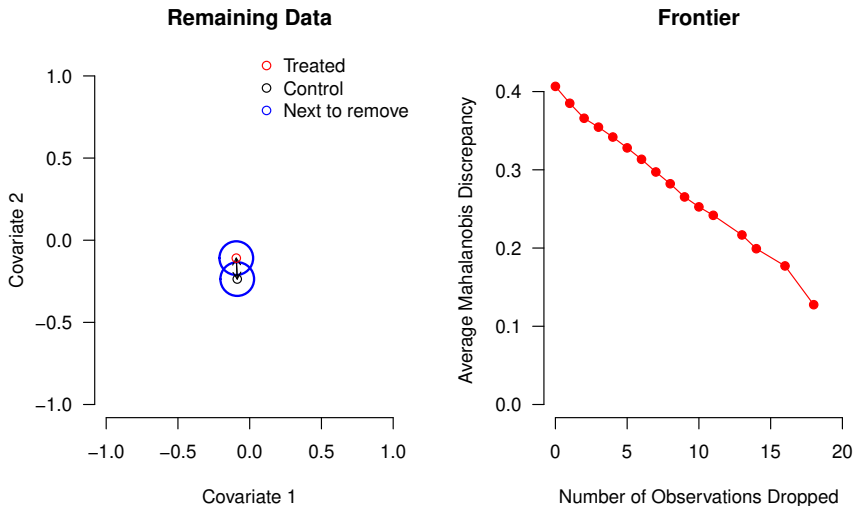


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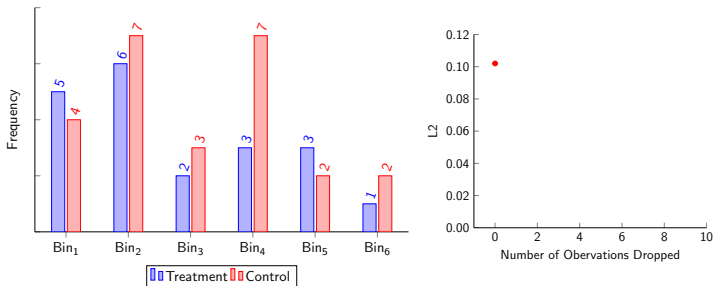
- Warning: figure omits details and the proof!

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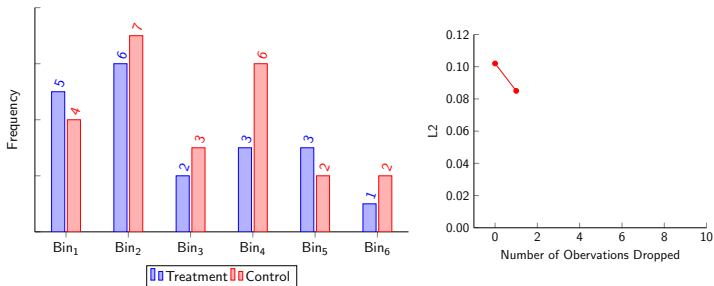


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- Very fast; works with any continuous imbalance metric

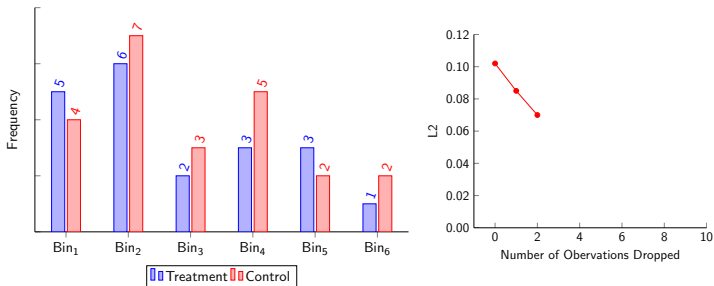
# Constructing the L1/L2 SATT Frontier



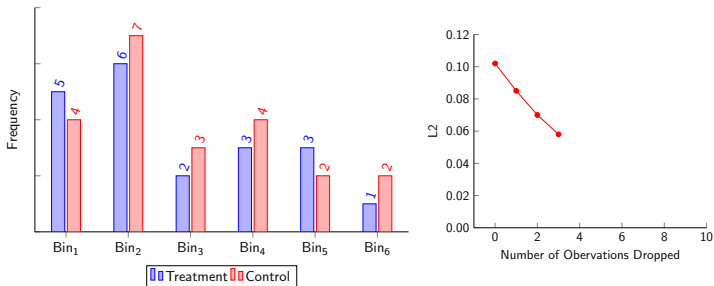
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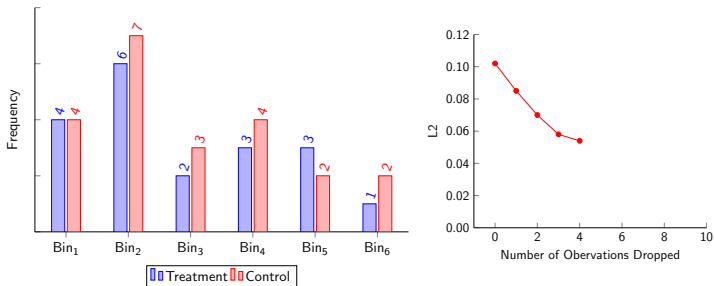
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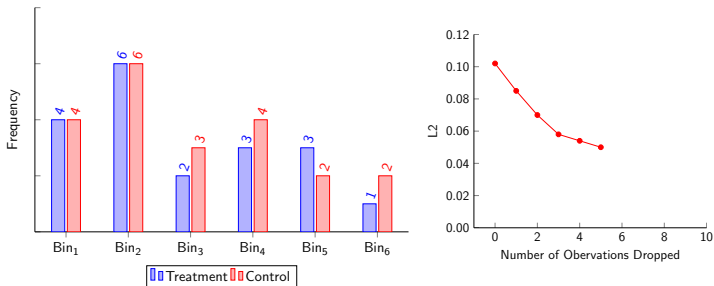
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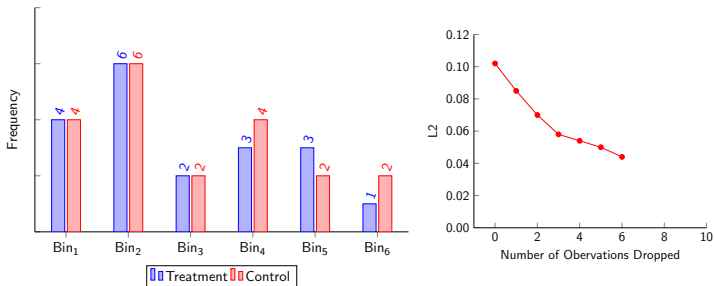
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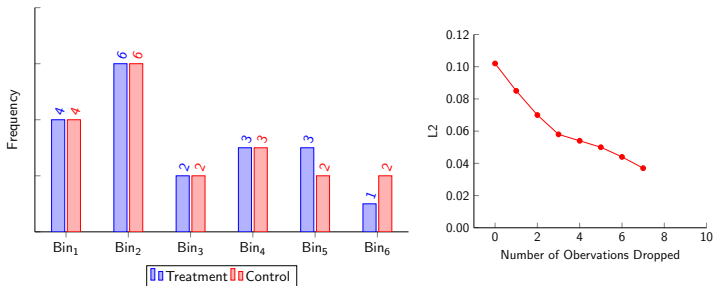
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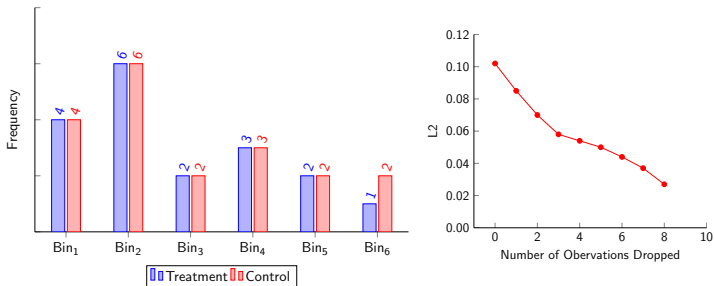
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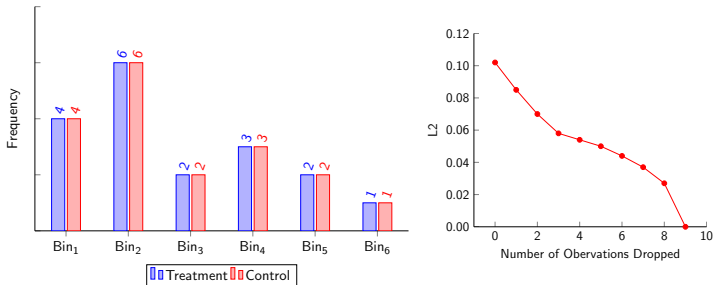
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- **↪ Using more information is simpler and more powerful**

For more information, papers, & software

GaryKing.org